

Aegivex AI — Feature Specification

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Prepared For: Development Team | **Prepared By:** Aegivex AI Team

1. Overview

This document describes every major feature of Aegivex AI, including its purpose, workflow, inputs, outputs, validations, and expected behavior. The primary objective is to ensure every feature is implemented consistently across the platform and provides a seamless, secure user experience.

2. Core Modules Summary

Feature	Primary Input	Key Checks / Purpose
AI Security Chat	User Prompt (max 2,000 chars)	Natural language interaction with AI Copilot for blockchain analysis.
Wallet Risk Scanner	Wallet Address	Reputation, suspicious activity, transaction history, blacklist status.
Token Risk Analyzer	Token Contract Address	Metadata, liquidity, honeypot detection, mint permissions, rug pull signals.
Smart Contract Analyzer	Smart Contract Address	Contract verification, dangerous functions, ownership permissions, proxy detection.
Website Safety Scanner	Website URL	SSL validity, domain age, domain reputation, phishing indicators.

Feature	Primary Input	Key Checks / Purpose
Transaction Explainer	Transaction Hash	Transaction type, requested permissions, token approvals, suspicious behavior.

3. Detailed Feature Specifications

3.1 AI Security Chat

Purpose: Allow users to communicate with the AI Security Copilot using natural language.

Inputs: User prompt

Outputs: AI response, Risk score (if applicable), Security explanation, Recommended actions

Workflow:

1. User enters a prompt.
2. Backend sends prompt to AI engine.
3. AI analyzes relevant blockchain context.
4. Response is generated with security insights.
5. Response is displayed in chat UI and saved to session history.

Validation: Prompt cannot be empty; maximum prompt length is 2,000 characters.

3.2 Wallet Risk Scanner

Purpose: Analyze the security profile and reputation of a blockchain wallet.

Input: Wallet Address

Checks: Wallet reputation, suspicious activity, transaction behavior, blacklist status

Outputs: Risk Score, Risk Level, Wallet Summary, Recommendation

Validation: Must be a valid wallet format on a supported blockchain network.

3.3 Token Risk Analyzer

Purpose: Evaluate whether a token contract is safe for interaction or investment.

Input: Token Contract Address

Checks: Token metadata, liquidity, ownership parameters, honeypot detection, mint permissions, rug pull indicators

Outputs: Risk Score, Token Summary, Security Recommendation

Validation: Must be a valid smart contract address.

3.4 Smart Contract Analyzer

Purpose: Review smart contract security attributes and risk exposure.

Input: Smart Contract Address

Checks: Contract verification status, dangerous functions, ownership permissions, proxy detection

Outputs: Security Summary, Risk Score, Recommendation

Validation: Must be on a supported network and have a valid contract address format.

3.5 Website Safety Scanner

Purpose: Detect phishing websites and malicious Web3 portals.

Input: Website URL

Checks: SSL certificate validity, domain age, web reputation, phishing indicators

Outputs: Trust Score, Risk Level, Security Recommendation

Validation: Must be a valid HTTP or HTTPS URL.

3.6 Transaction Explainer

Purpose: Decode and explain blockchain transactions before users sign or approve them.

Input: Transaction Hash

Checks: Transaction type, token approvals, requested permissions, suspicious activity

Outputs: Human-readable explanation, Risk Score, Warning messages, Recommended action

Validation: Must be a valid transaction hash.

4. Platform Infrastructure & Management

4.1 Dashboard

Purpose: Provide an intuitive overview of user security activities and threat metrics.

- **Displays:** Security Score, recent scans, threat alerts, scan statistics, AI recommendations.
- **Quick Actions:** Start Scan, Open AI Chat, View History.

4.2 Scan History

Purpose: Enable users to review, filter, and export previous security analyses.

- **Functions:** Search, Filter, Sort, Delete, Export.
- **Stored Data:** Scan Type, Risk Score, Date, Summary.

4.3 Notifications

Purpose: Keep users alerted on critical platform events and safety updates.

- **Types:** High Risk Alert, Scan Completed, Security Update, System Announcement.

4.4 Authentication & User Settings

Authentication Functions: Register, Login, Logout, Password Reset, Profile Update.

Security Requirements: JWT Authentication, Password Hashing, Session Management.

User Settings Features: Profile Management, Theme Selection, Notification Preferences, Password Change, Account Management.

5. Non-Functional Requirements & Error Handling

The application backend and frontend must adhere to the following resiliency rules:

- Display clear, meaningful error messages to the end user.
- Handle invalid or malformed inputs gracefully without crashing.
- Implement exponential backoff retry logic for temporary AI service failures.
- Log unexpected errors centrally for monitoring.
- Ensure application state recovery to prevent unexpected downtime.

6. Roadmap & Future Features

- Browser Extension (real-time transaction interception)
- Mobile Application (iOS & Android)
- Continuous Wallet Monitoring
- AI Threat Prediction Engine
- Cross-Chain Security Analysis
- Enterprise Security Dashboard
- Public & Developer API Access

7. Acceptance Criteria

A feature is considered complete and ready for deployment when:

- All specified inputs pass strict validation checks.
- Expected analytical outputs and security scores are accurate and generated.
- Errors are handled gracefully according to specifications.
- UI/UX updates reflect scan results correctly.
- Data is safely stored in the backend repository.
- Performance and latency meet hackathon quality metrics.

8. Expected Outcome

All features work together to deliver a complete AI-powered Web3 Security Copilot that enables users to analyze wallets, tokens, smart contracts, websites, and blockchain transactions through an intuitive interface with clear AI-driven security insights.